

Another reason is the poor security measures implemented on smart devices.

Types of DDoS

The DDoS attacks can be broadly classified into two major categories – application layer attacks and protocol attacks.

Application layer attacks are launched using the application protocols such as HTTP and FTP, and since the OSI network mode puts these services on the seventh layer, they are also referred to as Layer-7 attacks. A DDoS on this layer can cause a two-fold impact on the target services by first flooding the available bandwidth, and second, by causing the server to consume computing resources serving the requests. This could be a simple URL request that is being sent to a Web server from a large number of sources. Legitimate users may then get a slow or no response at all. We will look at using the *mod_evasive* module for Apache Web Server to mitigate this kind of attack, later in this article.

The **protocol layer attacks** target the session (Layer 5) and transport (Layer 4) layer defined in the OSI network model. The attacks are typically manifested in ways that exploit the weaknesses in the protocol layer to exhaust the available connection limits. An example of such an attack is when TCP packets are sent to the target with a spoofed source address during the TCP handshake process. A spoofed source address causes the target system to send a response and wait for the next step to take place, which never happens. This causes the TCP layer to exhaust the number of connections available and prevents it from serving requests queued behind the spurious requests.

These attacks could be routed through the botnets. Since each bot in the botnet is a legitimate device, discerning a pattern in the incoming traffic could be next to impossible.

An overview of the *mod_evasive* module

Apache *mod_evasive* is a module developed by Jonathan Zdziarski under the GPL v2.0 license, which permits commercial use as well as distribution, on the condition that you attribute the licence and share the source code of any derivative work. The source code for *mod_evasive* is available on GitHub at https://github.com/jzdzarski/mod_evasive.

mod_evasive is a security module for Apache and provides you with a way to take evasive actions in case your Web server running Apache comes under a DDoS or a brute force attack. A brute force attack differs from a DDoS attack in the sense that it is commonly used to attempt logging in to a service using many combinations of user names and passwords. The combinations can run into the thousands.

The common thread between a DDoS attack and brute force attack is the volume of connections hitting your Web servers, and that's where the *mod_evasive* module comes into the picture.

It gets instantiated for each listener in Apache and, hence, provides the same scaling capabilities that come with the Apache Web server. The detection is performed by building an internal dynamic hash table consisting of IP addresses and URIs. When a request arrives, it does the following (from the official documentation available at https://github.com/jzdzarski/mod_evasive):

- The IP address of the requester is looked up on the temporary blacklist.
- The IP address of the requester and the URI are both hashed into a 'key'. A lookup is performed in the listener's internal hash table to determine if the same host has requested this page more than once within the past one second.
- The IP address of the requester is hashed into a 'key'. A lookup is performed in the listener's internal hash table to determine if the same

host has requested more than 50 objects within the past second (from the same child).

If any of the above returns true, the request is blocked and HTTP status code 403 (Forbidden) is sent to the client. In addition, you can also configure this module to run a system command and/or send an email notification when this happens for further actions, like logging a ticket, tweaking a firewall setting by using IPTables, or sending an email to the security teams.

Securing the Apache server using *mod_evasive*

Let's jump to some hands-on work with this module. The following steps will show you how to install and configure this module on an Apache 2.x server running on Ubuntu 18.04. Your steps may vary, depending on the distributions and versions you're using.

1. Install Apache 2.x and configure it to run at system startup:

```
sudo apt update
sudo apt install apache2
sudo systemctl status apache2
```

2. The default install of Apache 2.x server doesn't come with *mod_evasive*. You can install the module using the following command:

```
sudo apt install libapache2-mod-evasive
```

You will be prompted to verify the mailing configuration on the host so that the module can send notifications when a DDoS is detected. The installation will also enable the module and you can verify the same, using the following command:

```
a2query -m | grep evasive
```

3. The default configuration of *mod_evasive* comes with a very restrictive configuration. While we will explore the details of the